Institute of Physics, Chinese Academy of Sciences

Charge-Density-Wave Transitions of Dirac Fermions Coupled to Phonons

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1 Holstein Model

Microscopic model for electron-phonon coupling

$$H_{el} = -t \sum_{<i,j>,\sigma} (c^{\dagger}_{i,\sigma} c_{j,\sigma} + h.c.) - \mu \sum_{i,\sigma} \hat{n}_{i,\sigma}$$

$$H_{ph} = \sum_{i} \left(\frac{M\Omega^2}{2} \hat{X}_i^2 + \frac{1}{2M} \hat{P}_i^2 \right)$$

$$\hat{P}_i = \frac{\hat{X}_{i,l+1} - \hat{X}_{i,l}}{\Delta\tau}$$

$$H_{int} = g \sum_{i,\sigma} \hat{n}_{i,\sigma} \hat{X}_i$$

$$\lambda = g^2/(M\Omega^2 W)$$

$$U_{eff}(\omega) = -\frac{\lambda W}{1 - (\omega/\Omega)^2}$$

Most simplest models depicting electron-phonon physics

T. Holstein, Annals of Physics 8, 325 (1959)

1 Holstein Model

Electron phonon coupling

→ metal insulator transition

→ charge density wave & superconductivity

→ interplay with el-el interaction

Numerical status:

- Lang-Firsov transformation, 1D QMC solved
- 2D CT-INT QMC (weak couplings)
- 2D DQMC + SLMC

QMC Simulations:

No Sign Problem, dope it as you like

Autocorrelation Problem -> Unreliable Results

1 Why Dirac Fermions

Why turn to honeycomb lattice and Dirac fermions

Experimental side:

- Graphene: wonderful platform.
- Magic Angle Twisted Bilayer Graphene

Theoretical side:

- surface state of topological insulator
- mass generation under interaction

Dirac fermion + electron-phonon interaction

2 DQMC for Holstein Model

Partition function and Trotter decomposition

$$Z = Tr\{e^{-\beta H}\} = Tr\{e^{-\Delta\tau H_{int}} e^{-\Delta\tau K}\}^L + O((\Delta\tau)^2)$$

Integrating out the fermion degree of freedom gives a determinant

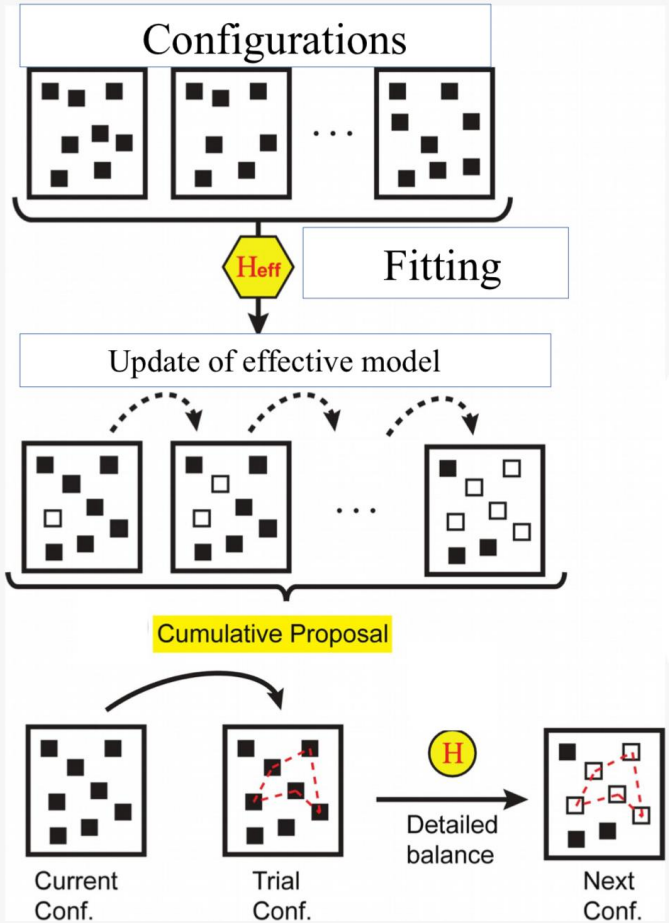
$$Z = \int dX e^{-E_{ph}\Delta\tau} det M_\uparrow det M_\downarrow$$

$$M_\sigma = I + B_L^\sigma B_{L-1}^\sigma \cdots B_1^\sigma$$

$$B_l^{\uparrow(\downarrow)} = e^{-\Delta\tau g X(l)} e^{-\Delta\tau K}$$

$$E_{ph} = \sum_{i,l} \frac{M\Omega^2}{2} X_{i,l}^2 + \sum_{i,l} \frac{M}{2} (\frac{X_{i,l+1} - X_{i,l}}{\Delta\tau})^2$$

2 Brief Intro on SLMC Method



$$Z = \sum_C \phi(C)\ \det\ (\mathbf{1} + \mathbf{B}(\beta, 0; C))$$

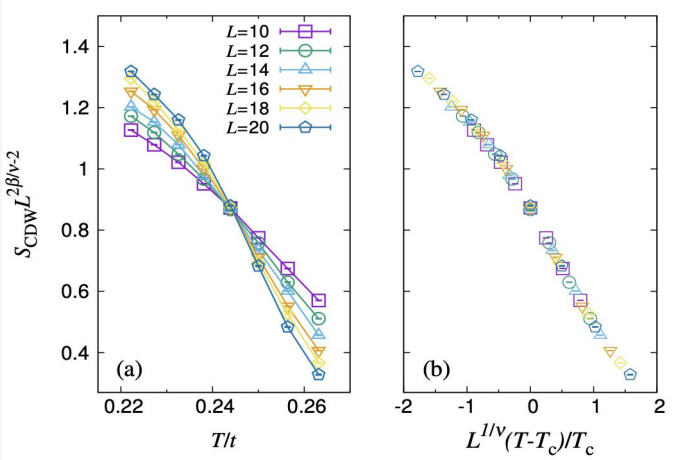
$$-\beta H^{\text{eff}}[C] = \ \ln\ (\omega[C])$$

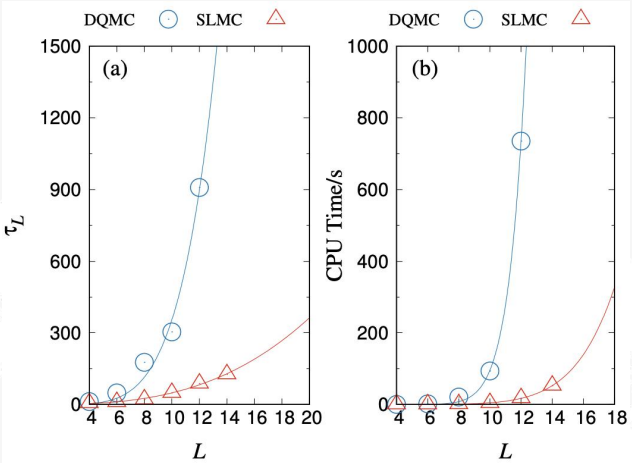
$$-\beta H^{\text{eff}} = J_k \sum_{i\tau}(X_{i\tau+1} - X_{i\tau})^2 + J_p \sum_{i\tau}\left(\frac{1}{4}(X_{i\tau} - \alpha)^4 - \frac{\alpha^2}{2}(X_{i\tau} - \alpha)^2\right) + J'_p \sum_{i\tau}\left(\frac{1}{6}(X_{i\tau} - \alpha)^6 - \frac{\alpha^2}{4}(X_{i\tau} - \alpha)^4\right)$$
$$+ J_{nn} \sum_{\langle ij\rangle\tau}(X_{i\tau} - \alpha)(X_{j\tau} - \alpha) + J'_{nn} \sum_{i\langle\tau\tau'\rangle}(X_{i\tau} - \alpha)(X_{i\tau'} - \alpha)\,. \qquad (1)$$

The effective model: PRB 98.041102

“efficient” update possible!

2 SLMC Holstein

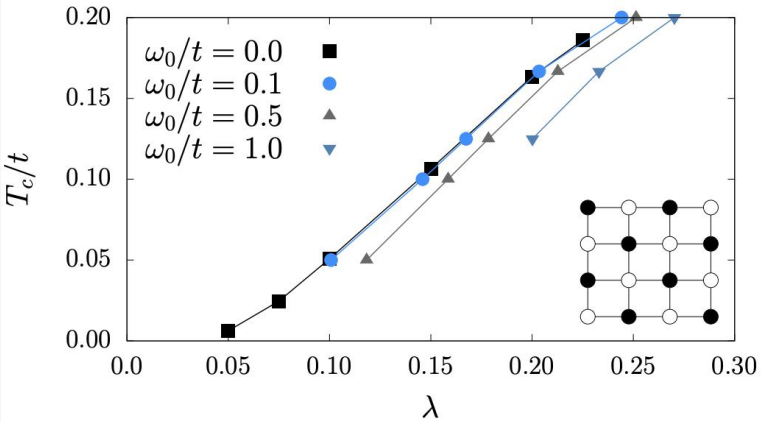


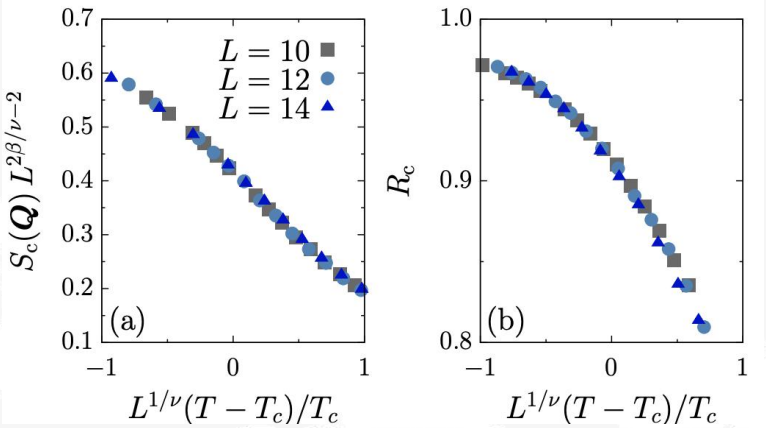


phonon frequencies and interactions hard for pure DQMC simulation

C. Chen, X. Y. Xu, J. Liu, G. Batrouni, R. Scalettar, and Z. Y. Meng, Phys. Rev. B 98, 041102

2 CT-INT Holstein

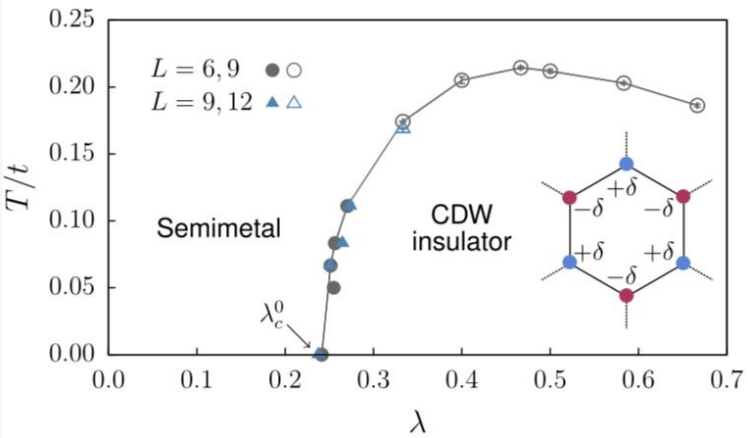




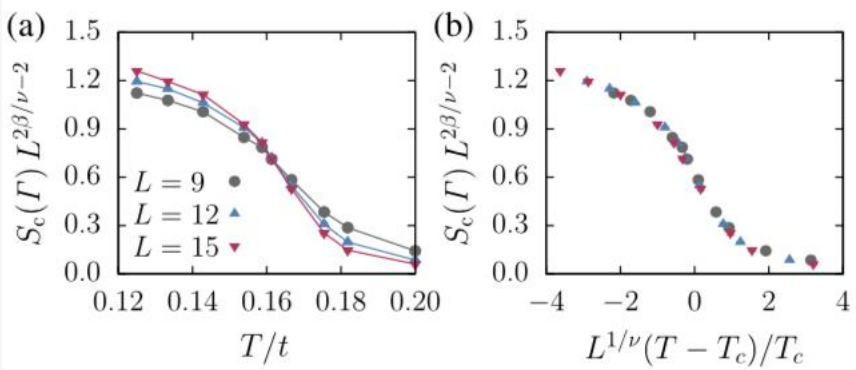
small phonon frequencies and interactions
reach very low temperatures
scheme: DQMC+SLMC phase diagram and CT-INT QPT and QCP

Square lattice Holstein-Hubbard model: M. Weber and M. Hohenadler, Phys. Rev. B 98, 085405

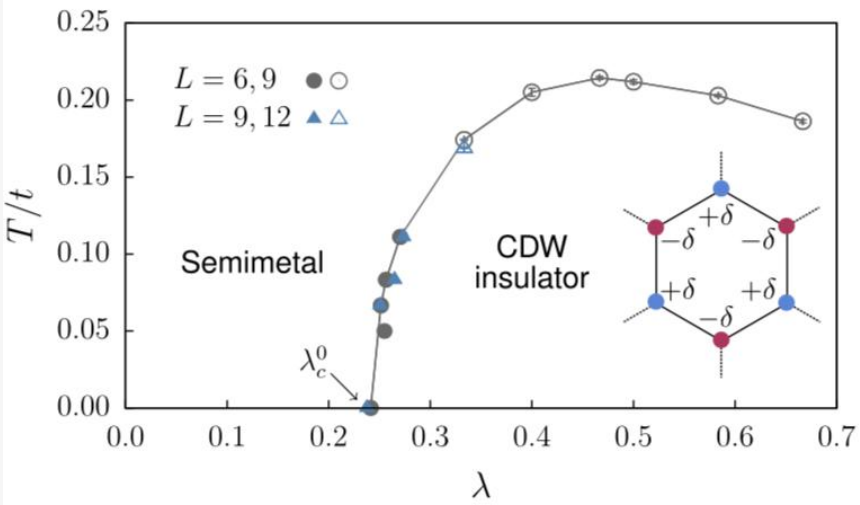
2 Results: Phase Diagram



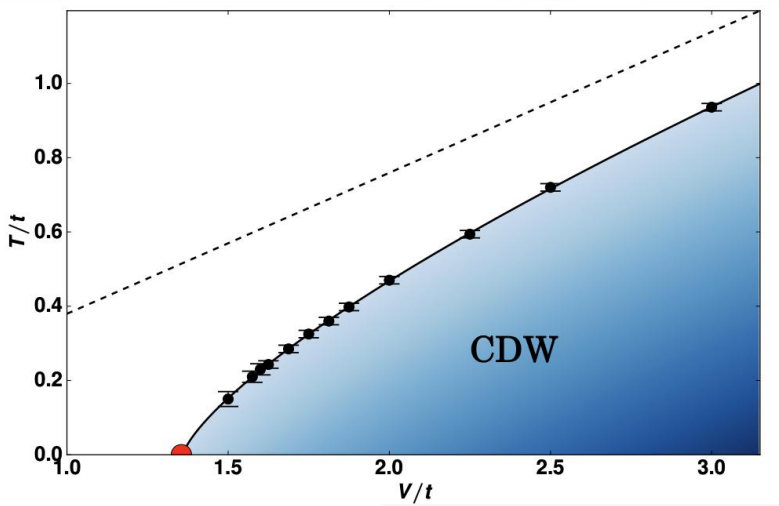
$$\omega_0 = 0.5$$



2 Results: Phase Diagram

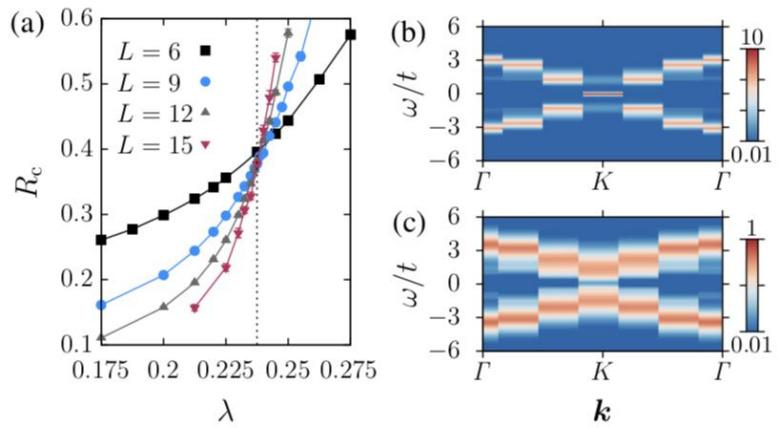


Spinless t-V model on honeycomb lattice



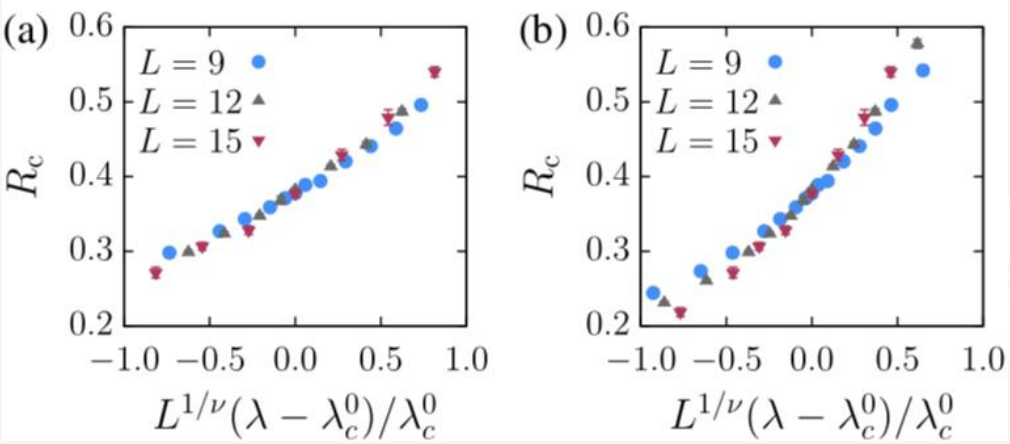
S. Hesselmann and S. Wessel, Phys. Rev. B 93, 155157

2 Results: Quantum Phase Transition



$1/\nu = 1.2$

$1/\nu = 0.931$



N. Zerf, L. N. Mihaila, P. Marquard, I. F. Herbut, M. M. Scherer, Phys. Rev. D 96, 096010

$N = 2$	$1/\nu$	η_ϕ	η_ψ	ω
this work, $P_{[2/2]}$	0.931	0.7079	0.0539	0.794
this work, $P_{[3/1]}$	0.945	0.6906	0.0506	0.777
$(2+\epsilon)$, (ϵ^4, Padé)[42]	0.931	0.745	0.082	
FRG[39]	0.994(2)	0.7765	0.0276	
conformal bootstrap[36]	0.88	0.742	0.044	
Monte Carlo[28]	1.20(1)	0.62(1)	0.38(1)	

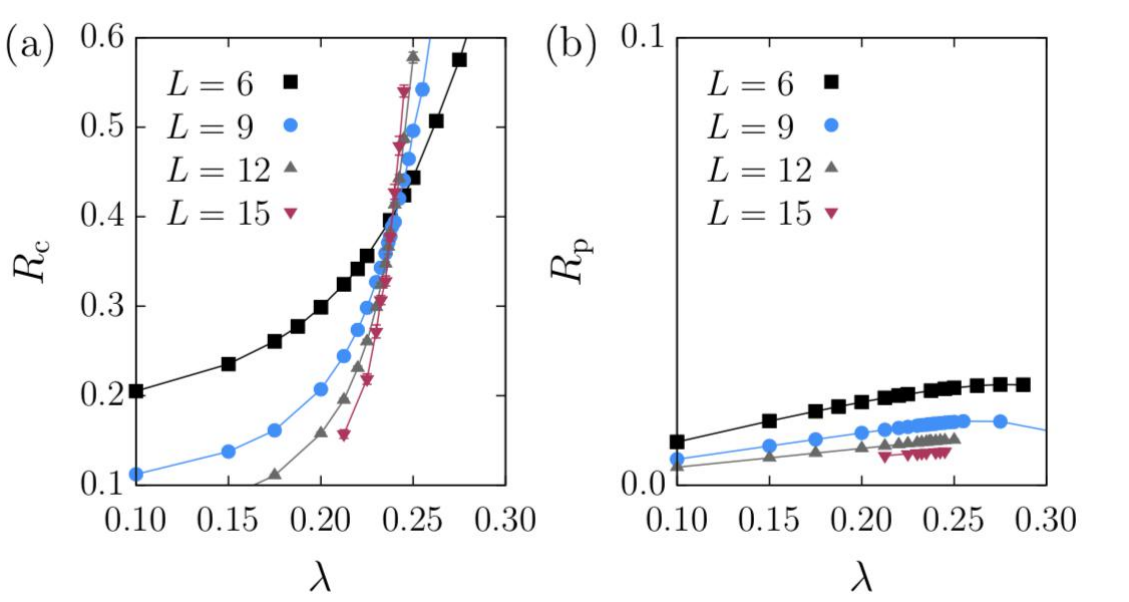
3 Summary

The Holstein Model on honeycomb lattice

- Phase diagram: non monotonic phase curve
- Quantum phase transition: in favor of chiral-Ising universality class
- Further direction: electron dispersion, doping, interplay with electron-electron interaction, topological phases to CDW transtions

Starting points: advancement of numerical methods + experiences

4 CDW vs SC


(a)
0.6
0.5
0.4
0.3
0.2
0.1
R_c
$L = 6$
$L = 9$
$L = 12$
$L = 15$
0.10
0.15
0.20
0.25
0.30
λ
(b)
0.1
0.0
R_p
$L = 6$
$L = 9$
$L = 12$
$L = 15$
0.10
0.15
0.20
0.25
0.30
λ